MD. Mehedi Hasan Maruf

Software Engineer

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Summary

Computer Science engineer (RUET) with almost 4 years of industry experience across full-stack web development (TypeScript, Python, Node.js) and hands-on .NET desktop development. Built a WPF-based pharmacy management system (.NET 8, Entity Framework Core, MVVM) alongside shipping Django/Node.js backends serving thousands of daily users, with practical DevOps experience in Docker, CI/CD, and Linux VPS management.

Skills

Programming Languages: Python, JavaScript, TypeScript
Web Technologies: Node.js, NestJS, Express.js, React.js, Angular, Next.js, FastAPI, Django, .NET
Databases: MySQL, PostgreSQL, MongoDB, Supabase
Tools & Concepts: Git, Docker, CI/CD (GitHub Actions, Netlify), VPS (Linux) Management, REST APIs, ORM (Django ORM, Supabase), Pydantic, Project Management, Problem Solving
AI/ML: LangChain, Hugging Face Transformers, Semantic Analysis, NLP

Experience

Software Engineer II

Jan 2025 – Present

Aamira Software Solutions, Kokomo, Indiana, USA

- Led development of a customer relationship management system that increased client response rates by 30%.
- Architected and implemented a Typescript microservice backend solution that reduced API response time by 40%.
- Optimized database queries using Supabase, resulting in 60% improvement in application performance.

Industrial Trainee

Mar 2025

Brain Station 23, Dhaka, Bangladesh

- Gained practical experience in enterprise software development methodology and practices.
- Served as Team Representative for a 15-person cohort, coordinating communication between management and trainees.
- Contributed to a student AI system using Next.js and Node.js that was deployed to an online server.

Junior Software Engineer

Nov 2022 – Apr 2024

Code Studio, Rajshahi, Bangladesh

- Developed backends supporting thousands of daily users using JavaScript, TypeScript, and Python (Django).
- Managed 12 VPS (Linux) with 99.8% uptime and optimized CI/CD pipelines, reducing deployment time by 40%.
- Implemented Docker containerization, simplifying deployment across 5 different client environments.

Personal Projects

StudentAI - Intelligent Learning Companion

Mar 2025

Read More:

- Developed during industrial attachment at Brain Station 23: AI-powered learning assistant combining **FastAPI backend** with **Next.js frontend** for personalized education.
- Implemented a **RAG pipeline** with LangChain for document Q&A, semantic retrieval, and conversation memory; extended to YouTube content analysis and intelligent quiz generation.
- Built a **multimodal** document processing pipeline supporting PDFs, images, and videos with real-time WebSocket chat, JWT authentication, and progress tracking.
- Deployed production-ready application on Vercel (frontend) and Render (backend) with Supabase database integration.

PharmaC - Desktop Pharmacy Management System *Feb 2025*

Read More:

- Built a **WPF desktop application** (.NET 8, MVVM via CommunityToolkit.Mvvm) for pharmacy inventory, sales, and order management, styled with the **WPF-UI** Fluent Design library.
- Designed a relational data layer with **Entity Framework Core** across **SQL Server** and **MySQL** (Pomelo) providers, modeling medicines, drug/herbal brands and generics, indications, and multi-role (admin/sub-admin) accounts.
- Implemented **JWT-based authentication** with persisted token refresh, barcode generation/lookup for medicine scanning, and a hardware-fingerprint device-licensing system (CPU/motherboard/BIOS hashing) for per-device activation.
- Integrated a REST API client (`HttpClient`) for medicine, order, cart, and license endpoints, enabling real-time inventory sync and localized (BDT) point-of-sale checkout.

Education

B.Sc. Engineering, Computer Science and Engineering *Mar 2022 – Aug 2026*

Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh

- CGPA: 3.75

Certifications & Achievements

- Digital Accessibility Training (DAT) - a2i (Aspire to Innovate), Govt. of Bangladesh (Oct 2025)
- Industrial Training Certificate - Brain Station 23 (Mar 2025)
- PHP (Laravel) - EDGE Project (Feb 2025)
- Experience Certificate - Code Studio (Apr 2024)
- 4th Place - RUET CodeSmash (Team Competition) - Competitive Programming Contest (Mar 2023)
- 4th Place - RUET CodeSpark (Individual) - Competitive Programming Contest (Oct 2022)

References

Dr. Md. Shahid Uz Zaman
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Profile

Dr. Md. Nour Hossain
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